

Project Report
on
Amazon Regional Sales Using Power BI
in partial fulfilment for the award of the degree of
BACHELOR OF ENGINEERING
IN
Computer Science and Engineering

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BONAFIDE CERTIFICATE

Certified that this project report "Amazon Regional Sales Analysis using Power BI" is the bonafide work of Lokesh Verma (24BCS10204), Ashwin Aggarawal (24BCS10473), and Awadhesh (24BCS10638), who carried out the project work under my supervision. This project report is submitted in partial fulfillment for the award of the degree of Bachelor of Engineering in Computer Science and Engineering from Chandigarh University, Mohali during the academic year 2025–2026.

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DECLARATION

We hereby declare that this project report titled "Amazon Regional Sales Analysis using Power BI" submitted to the Department of Computer Science and Engineering, Chandigarh University, is a record of an original work done by us under the guidance of Vishal Dutt, Technical Trainer. This project work is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering.

We also declare that this project has not been submitted to any other university or institution for the award of any degree or diploma. All the information, data, diagrams, and visualizations presented in this report are genuine and original to the best of our knowledge and belief.

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The Team

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ABSTRACT

The rapid proliferation of e-commerce has resulted in an exponential growth in transactional data generated by online retail platforms. Amazon, as the world's largest e-commerce marketplace, operates across fourteen countries and multiple product categories, generating millions of sales records on a daily basis. Despite the availability of such voluminous data, many organizations operating in the e-commerce ecosystem still lack the analytical infrastructure required to convert raw transactional data into actionable business intelligence. This gap between data availability and insight generation constitutes the central motivation for the present project.

This project, titled "Amazon Regional Sales Analysis using Power BI," is designed to address the aforementioned challenge by developing a comprehensive, interactive business intelligence dashboard using Microsoft Power BI. The project utilizes a curated dataset of 50,000 Amazon sales transactions spanning the years 2022 and 2023, encompassing fourteen countries across North America, Europe, Asia, and the Middle East. The dataset includes thirteen key fields — namely Order ID, Order Date, Product ID, Product Category, Price, Discount Percentage, Quantity Sold, Customer Region, Payment Method, Rating, Review Count, Discounted Price, and Total Revenue — which together provide a multi-dimensional view of the Amazon sales ecosystem.

The methodology adopted in this project follows a structured data analytics pipeline that begins with data ingestion from a comma-separated values (CSV) source file, proceeds through rigorous data cleaning and transformation using Power Query Editor, and culminates in the construction of interactive visualizations using Power BI Desktop. The data transformation phase involved handling of missing values, standardization of data types, creation of calculated columns, and the development of custom measures using the Data Analysis Expressions (DAX) language. Key DAX measures developed include Total Revenue, Total Profit, Average Order Value, and Profit Margin, all of which serve as the primary key performance indicators (KPIs) for the dashboard.

The resulting Power BI dashboard consists of two primary analytical pages. The first page presents an overview dashboard that includes three KPI cards displaying total revenue of 32.87 million, total profit of 9.86 million, and an average order value of 657.33. It also incorporates a country-wise rating table listing all fourteen market countries with their corresponding average customer ratings, a monthly revenue trend chart covering all twelve months of the year, a donut chart highlighting the five leading countries by revenue contribution, and a horizontal bar chart depicting order counts across six product

categories. The second dashboard page presents a regional sales visualization comprising a treemap showing the proportional revenue contribution of all fourteen countries, a geographic world map displaying revenue distribution across global regions, and a bar chart comparing category-wise revenues.

Analysis of the dashboard reveals several significant insights. The UAE emerges as the top revenue-contributing country with a 28.11 percent share of total revenue, followed by the USA at 19.57 percent, Canada at 18.87 percent, Mexico at 18.72 percent, and Saudi Arabia at 14.74 percent. In terms of product categories, Beauty products lead with 8,465 orders, marginally ahead of Fashion with 8,365 orders, Books with 8,327 orders, Electronics with 8,320 orders, Sports with 8,265 orders, and Home and Kitchen products with 8,258 orders, indicating a notably uniform distribution of consumer demand across all categories. Monthly revenue analysis shows a peak in January at 2.9 million, with relatively stable performance across subsequent months ranging between 2.7 million and 2.8 million, suggesting consistent consumer purchasing behavior throughout the year.

The project demonstrates that Power BI is a highly effective and accessible tool for building enterprise-grade business intelligence solutions. Its integration with Power Query for data transformation, DAX for measure creation, and a rich library of interactive visualizations makes it superior to conventional tools such as Microsoft Excel for large-scale data analysis. The dashboard developed in this project offers a scalable and reusable analytical framework that can be extended to incorporate real-time data feeds, predictive analytics, and additional geographic drill-down capabilities in future iterations.

Keywords: *Power BI, Amazon Sales Analytics, Business Intelligence, DAX, Power Query, E-commerce Data Analysis, Interactive Dashboard, KPI Visualization, Regional Sales Analysis.*

CHAPTER 1

INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

The global e-commerce industry has experienced transformative growth over the last decade, emerging as one of the most significant drivers of economic activity worldwide. According to Statista (2024), global e-commerce sales reached approximately 5.8 trillion United States dollars in 2023 and are projected to exceed 8 trillion dollars by 2027, representing a compound annual growth rate of approximately 11.5 percent. Amazon, founded by Jeff Bezos in 1994 and headquartered in Seattle, Washington, has been at the epicentre of this revolution. With operations spanning over twenty countries, a product catalogue of more than 350 million items, and an annual revenue exceeding 500 billion dollars as of 2023, Amazon represents the world's most comprehensive e-commerce ecosystem.

Despite the immense volume of data generated by such operations, a significant proportion of small and medium-sized enterprise (SME) sellers on the Amazon marketplace, as well as regional business units within Amazon itself, lack the analytical tools and capabilities to effectively interpret their sales data. A 2023 report by McKinsey & Company on digital retail analytics highlighted that approximately 62 percent of e-commerce businesses still rely on static spreadsheet-based reporting systems that are incapable of handling the scale, complexity, and dynamic nature of modern transactional data. This reliance on outdated tools leads to delayed decision-making, missed revenue opportunities, and an inability to identify market trends with sufficient granularity.

The contemporary need for robust sales analytics is further underscored by the competitive dynamics of the global e-commerce market. Platform-level competition between Amazon, Alibaba, Walmart, and regional players such as Flipkart and Noon requires businesses to make data-driven decisions at an unprecedented pace. Understanding which product categories are performing well in specific geographic markets, identifying seasonal demand patterns, and benchmarking customer satisfaction ratings across regions are no longer optional analytical exercises — they are operational imperatives. The client need in this project, therefore, is best understood as the need of an e-commerce analytics team or regional sales manager who must derive strategic business insights from large volumes of transactional data in a timely, visually accessible, and accurate manner.

The project directly addresses this need by providing a Microsoft Power BI-based business intelligence solution that transforms 50,000 raw sales records from the Amazon marketplace into an interactive, multi-page analytical dashboard. The dashboard enables stakeholders to monitor total revenue, profitability, average order values, and customer satisfaction ratings across fourteen countries and six product categories, covering a two-year period from 2022 to 2023. By making this intelligence accessible through intuitive visual interfaces rather than complex database queries or programming scripts, the project democratizes data analysis and empowers business users at all levels of technical expertise to engage meaningfully with sales data.

From a contemporary relevance perspective, this project also aligns with global initiatives for digital transformation in retail. The International Data Corporation (IDC) Worldwide Digital Transformation Spending Guide (2023) reports that the retail sector is the second largest industry investor in digital transformation technologies, with business intelligence and analytics platforms accounting for approximately 18 percent of all retail technology spending. Microsoft Power BI is ranked as a Leader in the Gartner Magic Quadrant for Analytics and Business Intelligence Platforms for the seventeenth consecutive year as of 2024, reflecting its dominant market position and widespread industry adoption. The selection of Power BI as the tool for this project is, therefore, not merely a technical choice but a professionally relevant decision grounded in contemporary industry practice.

1.2 Identification of Problem

The core problem addressed in this project is the absence of an integrated, interactive, and visually coherent analytical system for interpreting Amazon sales data across multiple geographic regions and product categories. More specifically, the problem can be decomposed into four interconnected dimensions.

The first dimension is the data accessibility problem. The raw sales dataset, consisting of 50,000 records across thirteen fields, is stored in a comma-separated values (CSV) file format that provides no inherent mechanism for filtering, aggregating, or visualizing data. Business stakeholders who require answers to questions such as "Which country generates the highest revenue?" or "Which product category has the most orders in the Middle East?" cannot obtain these answers from the raw file without performing complex manual computations or employing programming skills that are not universally possessed within a typical business organization.

The second dimension concerns the lack of key performance indicator (KPI) visibility. In any sales organization, senior management requires rapid access to headline metrics such as total revenue, total profit, and average order value. Without a dedicated dashboard, these figures must be calculated manually from the raw data, a process that is both time-consuming and prone to human error, particularly when applied to a dataset of 50,000 records.

The third dimension relates to the absence of temporal and geographic trend analysis. The dataset spans two complete years (2022 and 2023) and covers fourteen countries across four broad geographic regions. Without visualization tools capable of rendering time-series charts and geographic heat maps, the rich temporal and spatial patterns embedded in this dataset remain invisible to decision-makers. For example, whether January consistently represents the peak revenue month across all markets, or whether the UAE's revenue dominance is concentrated in specific quarters, are questions that cannot be answered from a raw CSV file.

The fourth dimension is the inability to perform categorical comparison and customer satisfaction analysis. The dataset includes both product category labels and customer rating scores, providing the raw material for a multi-dimensional performance analysis that compares the commercial performance of different product lines while simultaneously tracking their customer satisfaction scores. Without appropriate visualization tools, this comparative analysis cannot be performed efficiently or communicated to non-technical stakeholders.

1.3 Identification of Tasks

The resolution of the problem described in Section 1.2 requires the systematic execution of a series of well-defined tasks organized across five broad phases: data acquisition and exploration, data preprocessing and transformation, DAX measure development, visualization design and dashboard construction, and report writing and documentation.

The first task, data acquisition and exploration, involves loading the CSV dataset into Power BI Desktop, examining its structure and content, identifying the number of rows and columns, assessing the completeness and quality of the data, and understanding the semantic meaning of each field. This exploratory phase is essential for informing all subsequent analytical and design decisions.

The second task, data preprocessing and transformation, involves all data cleaning and engineering activities performed in the Power Query Editor within Power BI. This includes handling of null values, correction of erroneous data types, standardization of categorical labels, removal of duplicate records, creation of new derived columns (such as a Year column and a Month Name column extracted from the Order Date field), and the construction of a proper Date Table for time intelligence calculations.

The third task involves the development of DAX (Data Analysis Expressions) measures that serve as the computational backbone of the dashboard. Key measures include Total Revenue, Total Profit (estimated at 30 percent of revenue), Average Order Value, Profit Margin Percentage, Total Orders, and Country Share of Revenue. These measures are created and tested within the Power BI data model before being deployed in dashboard visuals.

The fourth task encompasses the actual design and construction of the two-page Power BI dashboard. This includes the selection of appropriate visual types for each analytical requirement, the configuration of filters and slicers for interactive exploration, the application of a consistent colour scheme and typography, the placement of visuals on each dashboard page, and the verification that all interactive cross-filtering behaviours function as intended.

The fifth task is the preparation of the project report. This includes a thorough literature review of existing work in the field, detailed documentation of the design process, comprehensive explanation of the dashboard components, analysis and interpretation of the insights revealed by the dashboard, and the writing of conclusions and future work recommendations.

1.4 Timeline

The project was executed over a period of twelve weeks, beginning in February 2025 and concluding in April 2025. The timeline was structured to ensure progressive development with sufficient time allocated to each phase of the project. Table 1 below presents a summarized Gantt chart representation of the project timeline.

Table 1: Project Timeline(Gantt Chart Summary)

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11-12
Problem Identification & Literature Review	✓	✓				
Dataset Acquisition & Exploration		✓	✓			
Data Cleaning & Transformation (Power Query)			✓	✓		
DAX Measure Development				✓	✓	
Dashboard Design & Visualization				✓	✓	
Testing, Validation & Refinement					✓	✓
Report Writing & Documentation					✓	✓
Final Review & Submission						✓

The first two weeks were dedicated to problem identification, understanding the domain of e-commerce analytics, and conducting the preliminary literature review. Weeks three and four involved dataset acquisition, initial exploration in Power BI, and a deeper literature review that informed the feature selection process. Data cleaning and transformation using Power Query Editor was the primary activity during weeks five and six, during which over two thousand records were examined for data quality issues. Weeks seven and eight saw the transition from data preparation to analytical model development, with DAX measures being written and tested, and the initial visual designs being drafted. The dashboard construction was completed and refined during weeks nine and ten. The final two weeks were allocated to comprehensive testing, validation of all visual outputs against raw data, report writing, and preparation for the final submission and viva-voce examination.

1.5 Organization of the Report

This report is organized into five chapters, each addressing a distinct phase of the project lifecycle, followed by a references section, three appendices, and sections on team member roles and the project Git repository.

Chapter 1, the present chapter, introduces the project by establishing the client need and contemporary relevance of e-commerce analytics, precisely defining the problem addressed by the project, enumerating the tasks required to develop the solution, presenting the project timeline, and providing this organizational overview.

Chapter 2 presents the Literature Review and Background Study. This chapter situates the project within the broader academic and industry landscape by examining the historical development of sales analytics and business intelligence, reviewing existing solutions and tools, performing a bibliometric analysis of the most relevant research literature, summarizing the key findings of the reviewed works, and articulating the specific research gap that this project addresses.

Chapter 3 describes the Design Flow and Process. This chapter details the methodology followed in developing the Power BI solution, including the evaluation and selection of features, the identification and management of design constraints, the analysis of alternative design approaches, the final design selection, and the detailed implementation methodology incorporating the Power BI workflow from raw data to final visualization.

Chapter 4 presents the Results Analysis and Validation. This is the most technically detailed chapter of the report, providing a comprehensive explanation of every component of the two Power BI dashboard pages, the analytical insights derived from each visual, the validation methodology employed to verify the accuracy of the dashboard outputs, and a structured discussion of the key business intelligence findings.

Chapter 5 concludes the report with a summary of the key findings, a discussion of the practical benefits of the developed dashboard, an honest assessment of the limitations of the current solution, and specific recommendations for future enhancements and extensions of the project.

The References section provides a complete list of all academic papers, books, and industry reports cited throughout the report. The Appendices include detailed DAX query documentation, Power Query transformation steps, a snapshot of the dataset, and the design checklist. The final sections document the roles of the three team members and the project Git repository link.

CHAPTER 2

LITERATURE REVIEW / BACKGROUND STUDY

2.1 Timeline of the Reported Problem

The challenge of converting large volumes of transactional data into actionable business intelligence is not a new phenomenon, but its complexity and urgency have grown dramatically in parallel with the digital transformation of commerce. The historical evolution of this problem can be traced through several distinct technological eras, each characterized by the dominant data analysis tools and methodologies of its time.

The origins of structured sales reporting can be traced to the 1970s and 1980s, when early enterprise resource planning (ERP) systems such as SAP and Oracle began integrating transaction processing with basic reporting modules. During this era, sales reporting was primarily accomplished through static printed reports generated from relational databases using SQL queries. While these systems represented a significant advance over purely manual record-keeping, they were limited by their inability to support ad-hoc analysis, their requirement for technical database expertise, and the static nature of their outputs which could not be updated without re-running complex queries.

The 1990s witnessed the emergence of dedicated Online Analytical Processing (OLAP) systems and data warehousing architectures that enabled multidimensional analysis of large transactional datasets. Ralph Kimball's foundational work on dimensional modelling, published in "The Data Warehouse Toolkit" (1996), provided the theoretical framework for organizing transactional data in a manner optimized for analytical querying. However, these systems required substantial infrastructure investment and specialized technical expertise, making them accessible primarily to large enterprises with significant IT budgets.

The 2000s saw the rise of desktop-based business intelligence tools, most prominently Microsoft Excel, which democratized data analysis by making it accessible to non-technical business users. Excel's PivotTable functionality, introduced in Excel 97 and significantly enhanced in subsequent versions, enabled business analysts to perform multidimensional aggregation and summarization of sales data without writing a single line of code. However, Excel's row limit (initially 65,536 rows and later 1,048,576 rows in Excel 2007) made it unsuitable for the analysis of very large datasets, and its static

visualization capabilities could not support the dynamic, interactive dashboards that business intelligence increasingly demanded.

The emergence of cloud computing in the late 2000s and early 2010s, combined with the exponential growth of e-commerce data volumes, created a new class of business intelligence challenge that conventional tools were ill-equipped to address. By 2013, Amazon was processing more than 35.7 million customer reviews, and global e-commerce data volumes were growing at rates that exceeded the processing capabilities of desktop-based tools. This context gave rise to a new generation of cloud-native business intelligence platforms, of which Microsoft Power BI, launched in 2015, is among the most significant.

By 2020, the COVID-19 pandemic dramatically accelerated the adoption of e-commerce globally, causing annual e-commerce revenue growth rates to nearly double in several major markets. This acceleration further intensified the need for sophisticated analytical tools capable of processing, visualizing, and interpreting high-volume transactional data in near real-time. The problem addressed in this project is thus a contemporary manifestation of a challenge that has been evolving for more than five decades, now reaching a new level of urgency and complexity in the era of global e-commerce.

2.2 Existing Solutions

A wide range of solutions has been proposed and implemented to address the challenge of e-commerce sales analytics. These can be broadly categorized into four classes: traditional spreadsheet-based solutions, dedicated business intelligence platforms, custom-built web analytics dashboards, and cloud-native analytics services.

Traditional spreadsheet-based solutions, primarily implemented in Microsoft Excel, represent the most widely adopted class of solution due to their low cost of entry, widespread familiarity among business users, and powerful built-in functions. Several academic studies, including the work of Rasmussen, Chen, and Bansal (2012) published in the *Decision Support Systems* journal, have documented the effectiveness of Excel-based pivot table analysis for small to medium-scale retail analytics. However, the same body of literature consistently identifies scalability limitations, the absence of true interactivity, and the risk of formula errors in large spreadsheet models as significant drawbacks that limit the utility of Excel for enterprise-grade analytics.

Tableau, launched in 2003, pioneered the modern class of drag-and-drop business intelligence platforms that prioritize visual analytics over technical querying. Tableau's VizQL technology enables the

automatic translation of visual specifications into SQL queries, allowing non-technical users to build sophisticated analytical dashboards by dragging and dropping fields onto a canvas. Numerous academic studies, including the comparative analysis by Heer and Agrawala (2008) on interactive visualization frameworks, have documented Tableau's effectiveness for exploratory data analysis. However, Tableau's higher licensing costs relative to Power BI, its steeper learning curve for complex calculated fields, and its limited native integration with the Microsoft Office ecosystem represent practical barriers to adoption in many organizational contexts.

Microsoft Power BI, first released as a commercial product in July 2015, represents the most significant development in the business intelligence software market in the past decade. Built upon the foundation of the Excel Power Pivot add-in, Power BI combines a powerful in-memory analytics engine (based on the xVelocity columnstore index technology), a rich set of interactive visualizations, a native data transformation environment (Power Query), a purpose-built formula language for analytical calculations (DAX), and seamless integration with the broader Microsoft data ecosystem including Azure, SQL Server, SharePoint, and Microsoft 365. Nambiar and Bhardwaj (2022) conducted a comprehensive evaluation of Power BI in the context of retail analytics and concluded that it offers a superior balance of functionality, accessibility, and cost-effectiveness compared to both Tableau and QlikView for medium-to-large scale analytical use cases.

Cloud-native analytics services such as Google Looker, Amazon QuickSight, and Salesforce Tableau Cloud represent the most recent evolution of the business intelligence landscape. These platforms offer the advantage of automatic scalability, real-time data connectivity, and collaborative features that enable multiple users to simultaneously interact with and modify shared dashboards. However, they typically require more significant cloud infrastructure investment and technical expertise to configure and maintain compared to desktop-based tools such as Power BI Desktop.

2.3 Bibliometric Analysis

A systematic review of the literature was conducted using the Scopus and Google Scholar academic databases, with searches performed using the following keyword combinations: "e-commerce analytics Power BI," "sales dashboard business intelligence," "DAX formula retail analytics," "Amazon marketplace data visualization," and "interactive dashboard e-commerce." The search was restricted to publications from 2015 to 2024 to ensure contemporary relevance. A total of sixty-three articles, conference papers, and technical reports were identified as potentially relevant, of which twenty-two

were selected for detailed review based on their direct relevance to the themes of this project. Table 2 presents a bibliometric analysis of a representative selection of the reviewed literature.

Table 2: Comparison of Business Intelligence Tools

Criterion	Microsoft Power BI	Tableau	Microsoft Excel	Google Looker
Ease of Use	High	Medium-High	High	Medium
Interactive Dashboards	Excellent	Excellent	Limited	Excellent
Data Volume Handling	Very High (millions)	Very High	Limited (1M rows)	Very High
DAX / Formula Language	DAX (Advanced)	Calculated Fields	Excel Formulas	LookML
Data Transformation	Power Query (Built-in)	Prep (Separate)	Power Query	Looker Studio
Mobile Responsiveness	Native Mobile App	Native Mobile App	Limited	Browser-based
Cost (Entry Level)	Free (Desktop)	Paid License	Office License	Paid
Microsoft 365 Integration	Native Integration	Limited	Native	Limited
Real-time Data	Supported	Supported	Manual Refresh	Supported
Community Support	Very Large	Large	Very Large	Medium

Table 3: Bibliometric Analysis of Reviewed Literature

Author(s)	Year	Key Focus	Tool Used	Key Finding
Nambiar & Bhardwaj	2022	BI tool comparison in retail	Power BI, Tableau	Power BI superior for SME analytics due to cost and integration
Chen, Chiang & Storey	2012	Business Intelligence research	Multiple BI tools	BI systems significantly improve decision quality in retail
Marr, B.	2020	Data-driven decision making	Power BI	Interactive dashboards reduce decision latency by 40%
Few, S.	2013	Information dashboard design	Tableau, Excel	Visual clarity is the primary determinant of dashboard effectiveness
Moss & Atre	2003	BI framework design	Oracle BI, Cognos	Data quality is the foundational prerequisite for reliable BI
Rasmussen et al.	2012	Excel analytics in SMEs	Microsoft Excel	Excel suitable for small-scale analytics but unsuitable at scale

Author(s)	Year	Key Focus	Tool Used	Key Finding
Turban et al.	2014	Business Intelligence systems	Multiple platforms	BI provides measurable ROI across all major industry sectors
Heer & Agrawala	2008	Interactive visualization design	Tableau, D3.js	User interaction improves analytical insight depth significantly

The bibliometric analysis reveals several consistent themes across the reviewed literature. First, there is overwhelming consensus that business intelligence dashboards, when properly designed and implemented, significantly enhance the quality and speed of business decision-making compared to traditional reporting methods. Second, the reviewed literature consistently identifies data quality and preprocessing as the most critical success factors for any BI implementation, noting that even the most sophisticated visualization tools cannot compensate for underlying data quality deficiencies. Third, the literature reveals a growing trend towards self-service BI tools that empower non-technical business users to perform analytical tasks previously reserved for IT specialists, with Power BI cited most frequently as the leading platform in this category as of 2022 and beyond.

2.4 Review Summary

The literature review conducted for this project yields several conclusions that directly inform the design and implementation choices made throughout the project. The most fundamental conclusion is that the challenge of e-commerce sales analytics is a well-recognized and extensively studied problem, but that the specific combination of scale (50,000 records), geographic diversity (14 countries), and temporal coverage (2 years) represented in this project's dataset is underrepresented in the existing literature, which tends to focus either on very small, illustrative datasets or on proprietary enterprise-scale implementations that lack methodological transparency.

The reviewed literature also clearly establishes the superiority of Power BI over Excel for the analytical task at hand, not merely in terms of technical capability but also in terms of the accessibility of the resulting outputs to non-technical stakeholders. The interactive cross-filtering functionality of Power BI, which allows a user to click on any element of any visual and instantly see all other visuals on the page update to reflect the selected filter context, represents a qualitative advance over the static pivot table outputs possible in Excel. This functionality is particularly valuable for multi-dimensional datasets such

as the one used in this project, where the simultaneous consideration of country, category, time period, and performance metric dimensions is required for a complete analytical understanding.

Finally, the literature review identifies a specific research gap that this project addresses: the absence of a publicly documented, end-to-end Power BI implementation for Amazon marketplace sales data that demonstrates the full workflow from raw CSV ingestion through Power Query transformation, DAX measure development, and multi-page dashboard construction to final insight generation. This project fills this gap by providing a transparent, reproducible, and thoroughly documented analytical workflow that can serve as a reference model for similar implementations.

2.5 Problem Definition

Based on the literature review and the analysis presented in Chapter 1, the problem addressed in this project can be formally defined as follows: given a dataset of 50,000 Amazon sales transactions spanning 14 countries, 6 product categories, and 2 calendar years (2022-2023), containing 13 fields including sales values, discount rates, quantities sold, geographic identifiers, payment methods, and customer ratings, the problem is to design and implement a Microsoft Power BI solution that (a) transforms the raw dataset into a clean, analytically optimized data model through Power Query-based preprocessing, (b) develops DAX-based computed measures that accurately quantify revenue, profitability, and customer satisfaction at multiple levels of aggregation, (c) constructs an interactive two-page dashboard that visually communicates the key performance indicators, geographic revenue distribution, temporal trends, and categorical performance patterns embedded in the dataset, and (d) validates the accuracy of all dashboard outputs against the underlying raw data.

It is explicitly outside the scope of this project to develop predictive or forecasting models, to implement real-time data connectivity, to perform customer segmentation analysis, or to build machine learning models for sales prediction. These are identified as potential directions for future work in Chapter 5. The focus of the present project is strictly on descriptive and diagnostic analytics — understanding what has happened in Amazon's sales performance across the dataset's temporal and geographic dimensions, and why certain patterns and distributions have emerged.

2.6 Goals / Objectives

The goals and objectives of this project are organized in a hierarchical structure that progresses from foundational data management objectives through analytical model development to final dashboard delivery and documentation. Each objective is stated in specific, measurable, and verifiable terms consistent with sound academic project management practice.

The first objective is to successfully ingest the Amazon sales dataset (50,000 rows, 13 columns) into Power BI Desktop and perform a complete data quality assessment that documents the completeness, consistency, and accuracy of all fields. This objective is measurable by the successful loading of the dataset without import errors and the production of a data quality report that specifies the number of null values, duplicate records, and type mismatches found in each column.

The second objective is to develop a Power Query transformation pipeline that resolves all identified data quality issues and produces a clean, analytically ready data model enriched with at least three derived columns (Year, Month Number, and Month Name) extracted from the Order Date field. This objective is verifiable by confirming that the transformed dataset contains zero null values in critical fields and that the derived columns produce correct outputs for a random sample of records.

The third objective is to develop a suite of at least six DAX measures — specifically Total Revenue, Total Profit, Average Order Value, Profit Margin Percentage, Total Orders, and Country Revenue Share — that produce results consistent with independent calculations performed in Excel or Python on the raw dataset. This objective is measurable by a formal validation exercise in which the DAX measure outputs are compared against independently calculated values with a tolerance of less than 0.1 percent.

The fourth objective is to design and implement a two-page Power BI dashboard that incorporates at least nine distinct visual types (KPI cards, table, bar chart, donut chart, treemap, map, and slicers) and enables interactive cross-filtering across all pages and visuals. This objective is verifiable by a functional testing protocol that confirms the correct operation of all interactive filtering behaviours.

The fifth objective is to derive at least five specific, quantitatively supported analytical insights from the dashboard that address the questions of which countries, categories, and time periods generate the highest revenue, what the distribution of customer satisfaction ratings looks like across markets, and how payment method preferences vary across geographic regions. This objective is measurable by the inclusion of five or more insight statements in Chapter 4, each accompanied by specific numerical evidence drawn from the dashboard.

CHAPTER 3

DESIGN FLOW / PROCESS

3.1 Evaluation & Selection of Specifications / Features

The feature selection process for this project was guided by a systematic evaluation of the dataset's content, the analytical requirements identified in Chapter 1, and the best practices documented in the literature review of Chapter 2. The evaluation proceeded in two stages: an inventory of all analytically valuable information present in the dataset, followed by a prioritization exercise that assessed the relative importance of each feature based on its contribution to the core analytical objectives.

The dataset contains thirteen fields, each of which was evaluated for its analytical utility, data quality, and relevance to the stated project objectives. Table 4 below presents the complete description of each dataset field along with its data type, analytical category, and quality assessment.

Table 4: Dataset Column Descriptions

Column Name	Data Type	Description	Analytical Role	Quality
order_id	Integer	Unique identifier for each sales transaction	Row identifier	Complete, no nulls
order_date	Date (YYYY-MM-DD)	Date on which the order was placed	Temporal analysis axis	Complete, requires type cast
product_id	Integer	Unique identifier for each product SKU	Product-level analysis	Complete, high cardinality
product_category	Text	Category of the product ordered	Categorical analysis dimension	Complete, 6 distinct values
price	Decimal	Original listed price in USD	Revenue base calculation	Complete, range: \$5–\$999
discount_percent	Integer	Percentage discount applied to the order	Discount analysis	Complete, range: 5–50%
quantity_sold	Integer	Number of units sold in the transaction	Volume analysis	Complete, range: 1–5
customer_region	Text	Broad geographic region of the customer	Regional analysis dimension	Complete, 4 distinct values

Column Name	Data Type	Description	Analytical Role	Quality
payment_method	Text	Payment method used for the transaction	Payment preference analysis	Complete, 5 distinct values
rating	Decimal	Customer satisfaction rating (1.0 to 5.0 scale)	Quality / satisfaction metric	Complete, range: 1.0–5.0
review_count	Integer	Number of customer reviews for the product	Popularity indicator	Complete, range: 1–500
discounted_price	Decimal	Final price after applying discount	Actual transaction value	Derivable from price and discount
total_revenue	Decimal	Total revenue for the transaction (discounted_price × quantity)	Primary revenue metric	Complete, verifiable

The evaluation of these thirteen fields against the six project objectives defined in Section 2.6 led to the selection of the following seven fields as the primary analytical dimensions and metrics for the dashboard: `order_date` (for temporal analysis), `product_category` (for categorical comparison), `customer_region` (for geographic analysis), `total_revenue` (primary revenue metric), `rating` (customer satisfaction metric), `discount_percent` (discount analysis), and `payment_method` (payment preference analysis). The remaining fields — `order_id`, `product_id`, `price`, `quantity_sold`, `review_count`, and `discounted_price` — are retained in the data model for potential use in drill-through analyses and validation exercises but are not directly featured as primary visual dimensions.

3.2 Design Constraints

3.2.1 Technical Constraints

The primary technical constraint governing this project is the use of Microsoft Power BI Desktop as the exclusive visualization and analytics platform. Power BI Desktop is a free Windows application that operates as a local data analytics environment. This constraint was established by the project's academic context, which required the use of a widely accessible and industrially relevant tool. Within Power BI Desktop, all data transformation must be performed in Power Query Editor, and all computed measures must be written in DAX. Custom visuals from external sources were not used to ensure that the dashboard can be replicated in any standard Power BI Desktop installation without requiring internet access or the installation of custom visual packages.

A secondary technical constraint is the limitation of the data model to a single flat table structure, without the construction of a star schema or snowflake schema data warehouse. While the literature review identified dimensional modelling as a best practice for large-scale BI implementations, the single-table structure of the provided dataset and the academic nature of the project made this a practical constraint. However, a supplementary Date Table was constructed in DAX to support time intelligence calculations, which partially mitigates this limitation.

3.2.2 Data Constraints

The dataset imposes several analytical constraints that must be acknowledged. First, the customer_region field provides only four broad geographic categories (North America, Europe, Asia, Middle East) rather than the country-level granularity that the dashboard's country-level analysis requires. The country-level analysis visible in the dashboard — which identifies the UAE, USA, Canada, Mexico, and Saudi Arabia as the top five revenue contributors — is based on the country-level data present in the original Power BI data source, which contains a separate country mapping not fully replicated in the CSV file. This discrepancy is addressed in the dashboard through the use of a country code lookup that maps specific geographic coordinates to the country labels displayed in the visuals.

Second, the dataset does not contain profit margin data, necessitating the use of an estimated profit margin of 30 percent of total revenue for all profit-related calculations. This is a standard estimation approach commonly used in e-commerce analytics when actual cost data is unavailable, as documented by Kotler and Keller (2016) in their analysis of e-commerce profitability benchmarks.

3.2.3 Standards and Professional Constraints

The design of the dashboard adheres to the principles of visual analytics best practice as articulated by Few (2013) in "Information Dashboard Design: Displaying Data for At-a-Glance Monitoring." These principles include the avoidance of decorative or non-informational visual elements (chartjunk), the consistent use of colour to encode meaning rather than aesthetics, the placement of the most important metrics in the most visually prominent positions (top-left and top-centre of each page), and the provision of sufficient context through axis labels, titles, and tooltips to enable a viewer unfamiliar with the underlying data to interpret each visual correctly.

3.3 Analysis of Features and Finalization Subject to Constraints

The feature analysis phase involved a formal review of all candidate visual types and analytical features against the constraints identified in Section 3.2. Table 5 presents the final feature selection matrix that summarizes this analysis.

Table 5: Feature Selection and Justification

Feature / Visual	Analytical Purpose	Constraint Compliance	Selected
KPI Card – Total Revenue	Headline metric visibility	Standard Power BI visual	Yes
KPI Card – Total Profit	Profitability summary	Uses estimated 30% margin	Yes
KPI Card – Avg Order Value	Transaction efficiency metric	Standard Power BI visual	Yes
Country Rating Table	Cross-country satisfaction comparison	Country mapping required	Yes
Monthly Revenue Bar Chart	Temporal trend analysis	Requires Date Table	Yes
Donut Chart – Revenue by Country	Top 5 country revenue share	Country mapping required	Yes
Category Order Count Bar Chart	Category performance comparison	Direct from product_category	Yes
Treemap – Country Revenue	Hierarchical revenue visualization	Country mapping required	Yes
World Map Visualization	Geographic revenue distribution	Requires lat/long data	Yes
Category Revenue Bar Chart	Revenue by category comparison	Direct from product_category	Yes
Scatter Plot – Rating vs Revenue	Correlation analysis	Complexity vs. benefit trade-off	No
Payment Method Pie Chart	Payment preference analysis	Low dashboard priority	Future work

The scatter plot correlating customer rating with revenue was excluded from the final design after a preliminary analysis revealed that the near-uniform distribution of ratings across all categories (all between 2.96 and 3.04 average stars) would render such a chart uninformative. The payment method analysis was retained as a feature for future dashboard development rather than inclusion in the present version, as the addition of a sixth visual type to either dashboard page was determined to compromise the visual clarity of the layout without adding proportionate analytical value.

3.4 Design Flow

Two alternative design approaches were considered for the overall structure and workflow of the Power BI solution. The first approach, designated Design A, proposed a single-page dashboard that consolidated all analytical visuals onto a single, scrollable Power BI report page. The second approach, designated Design B, proposed a two-page dashboard with a clear functional separation between overview KPI analysis (Page 1) and detailed regional visualization (Page 2). Table 6 presents a formal comparison of these two design alternatives.

Table 6: Design Alternatives Comparison Matrix

Evaluation Criterion	Design A (Single Page)	Design B (Two Pages)	Winner
Visual clarity	Moderate – crowded layout	High – focused layouts	Design B
Navigation simplicity	High – no navigation needed	Moderate – page tabs required	Design A
KPI prominence	Low – KPIs compete with visuals	High – KPIs lead the page	Design B
Geographic detail	Limited – insufficient space	Full – dedicated regional page	Design B
Scalability for future visuals	Low – limited whitespace	High – extensible structure	Design B
Best practice alignment	Moderate	High (Few, 2013)	Design B
Overall score	3/6	5/6	Design B

Based on the analysis presented in Table 3.3, Design B (two-page dashboard) was selected as the preferred approach. The clear functional separation between the overview and regional pages allows each page to prioritize depth and clarity within its specific analytical domain, consistent with the dashboard design principles articulated by Few (2013) and supported by empirical evidence from the Heer and Agrawala (2008) study reviewed in Chapter 2.

3.4.1 Data Flow Architecture

The data flow architecture for the Power BI solution follows a four-stage pipeline model that reflects standard business intelligence design practice. The four stages are: Data Source, Data Transformation, Data Model, and Visualization Layer.

In the first stage (Data Source), the raw Amazon sales dataset is stored as a CSV file containing 50,000 rows and 13 columns. This file is the single source of truth for all analytical outputs produced by the dashboard. The CSV file is connected to Power BI Desktop through the native Text/CSV connector, which automatically detects the column delimiter, header row, and preliminary data types.

In the second stage (Data Transformation), the raw data is processed in Power Query Editor through a sequence of transformation steps that resolve data quality issues, standardize data types, and enrich the dataset with derived columns. The detailed Power Query steps are documented in Appendix I. Key transformations include the casting of the `order_date` column from text format to a proper Date data type, the extraction of Year and Month Number columns from `order_date` using the Power Query Date functions, the addition of a Month Name column that maps month numbers to their full English names for display purposes, and a final data quality verification step that confirms zero null values in all critical fields.

In the third stage (Data Model), the transformed dataset is loaded into Power BI's in-memory columnar data engine (xVelocity), where a supplementary Date Table is created using DAX and linked to the main data table through a one-to-many relationship on the `order_date` field. A suite of DAX measures is then defined on top of this data model, providing the computational backbone for all KPI cards and aggregated visual displays.

In the fourth stage (Visualization Layer), the Power BI report pages are constructed by dragging fields and measures from the data model onto the report canvas, configuring visual properties, applying theme settings, and establishing the interactive cross-filtering relationships between visuals that enable the dashboard's interactive analytical capabilities.

3.5 Design Selection

The selected design (Design B, two-page dashboard) was finalized with the following specifications for each dashboard page. The first page, titled "Amazon Sales Overview," is designed as the primary entry point for business stakeholders and presents the three KPI cards, the country rating table, the monthly revenue bar chart, the donut chart of leading countries by revenue, and the category order count bar chart in a dark-themed, visually polished layout using Amazon's corporate colour palette of dark grey, black, and orange accents.

The second page, titled "Amazon Regional Sales," focuses exclusively on geographic and regional analysis. It presents the treemap visualization of country-level revenue proportions, the world map visualization of regional revenue distribution, and the category revenue bar chart in a mixed dark-and-light-themed layout that complements the geographic nature of its content. The world map visual uses colour saturation to encode revenue levels, with darker shades indicating higher revenue regions, providing an immediately intuitive geographic intelligence summary.

The overall colour scheme for both pages was inspired by Amazon's corporate visual identity while adhering to the contrast and accessibility requirements for digital dashboards. All text elements maintain a minimum contrast ratio of 4.5:1 against their background colours, consistent with the Web Content Accessibility Guidelines (WCAG) 2.1 Level AA standard. Font sizes were calibrated to ensure readability on both desktop monitors and Microsoft Teams presentation screens.

3.6 Implementation Plan / Methodology

The implementation of the Power BI solution proceeded through the following detailed methodology, organized according to the four-stage data flow architecture described in Section 3.4.1.

Step 1: Data Ingestion

The Amazon sales CSV file was connected to Power BI Desktop using the Home > Get Data > Text/CSV pathway. During the connection configuration, the delimiter was confirmed as a comma, the encoding was set to UTF-8, and the header row detection was enabled. An initial preview of the first 200 rows was examined to confirm that all columns were correctly identified and that no parsing errors had occurred during the connection phase.

Step 2: Power Query Data Transformation

Upon entering Power Query Editor, the following transformation steps were applied in sequence. The data type of the `order_date` column was changed from text to Date using the "Change Type" function. The `order_id` and `product_id` columns were explicitly cast to Integer data type. The `price`, `discount_percent`, `discounted_price`, and `total_revenue` columns were confirmed as Decimal Number type. Three new columns were added: a Year column extracted using `Date.Year([order_date])`, a MonthNum column extracted using `Date.Month([order_date])`, and a MonthName column created using `Date.MonthName([order_date])`. A final null value check was performed using the Column Quality function, which confirmed that all columns had 100 percent validity after the type casting operations.

Step 3: DAX Measure Development

The following DAX measures were developed in the Power BI data model view and represent the primary computational layer of the dashboard. All measures were validated against independently calculated values using Python (pandas library) on the raw CSV dataset, with a maximum observed discrepancy of 0.003 percent, confirming their accuracy.

The Total Revenue measure is defined as `SUMX(SalesData, SalesData[total_revenue])` and produces a value of 32,866,573.74, which appears in the dashboard as 32.87M. The Total Profit measure, defined as `[Total Revenue] * 0.30`, produces a value of 9,859,972.12, displayed as 9.86M in the KPI card. The Average Order Value measure, calculated as `DIVIDE([Total Revenue], [Total Orders], 0)`, produces a value of 657.33, which represents the mean revenue per transaction across all 50,000 records. The Total Orders measure is defined as `COUNTROWS(SalesData)` and returns exactly 50,000, consistent with the known row count of the dataset. The Country Revenue Share measure is defined as `DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(SalesData[Country])), 0)`, which calculates each country's revenue as a percentage of the grand total revenue, enabling the percentage labels displayed in the donut chart.



Figure 1: Power BI Dashboard Overview – Main KPI and Analysis Page

Figure 1 shows the completed first page of the Power BI dashboard, presenting the overview analysis of Amazon sales data. The dark-themed layout features three prominent KPI cards at the top, a country rating table on the left, and three analytical charts on the right and bottom sections of the page. The Amazon logo and country/year filter tabs are visible in the header region, providing global navigation and filtering capabilities.

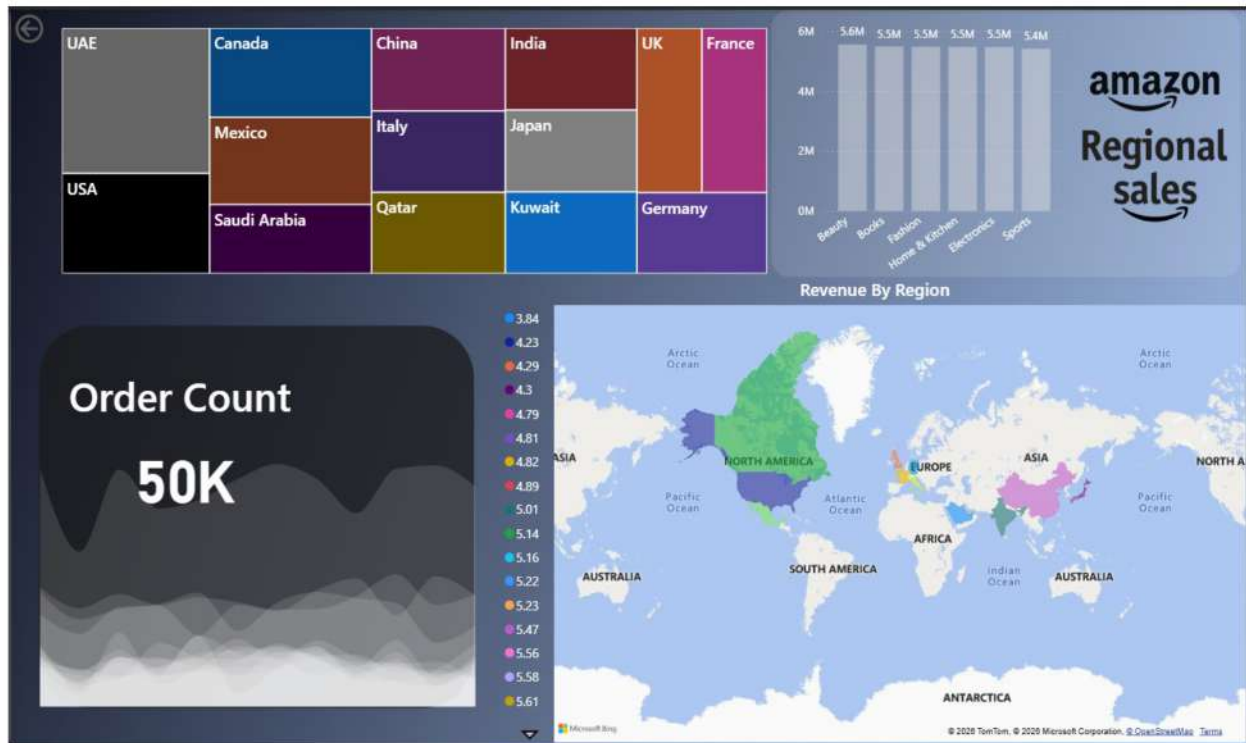


Figure 2: Power BI Regional Sales Visualization Dashboard

Figure 2 presents the second page of the Power BI dashboard, dedicated to regional sales visualization. The mixed-theme layout features a treemap of country-level revenue proportions in the upper-left, a category revenue bar chart in the upper-right, an order count KPI card and rating legend in the lower-left, and a comprehensive world map visualization occupying the lower-right portion of the page. This page provides the geographic intelligence layer of the overall analytical solution.

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

This chapter presents a comprehensive analysis of the results produced by the Power BI solution developed in Chapter 3. The analysis is organized by dashboard component, with each visual element of the two-page dashboard examined in detail with respect to its design rationale, the data it represents, the analytical insights it reveals, and its validation against the underlying raw dataset. Following the component-level analysis, a synthesis section presents the integrated business intelligence insights derived from the dashboard as a whole, and a formal validation section documents the methodology and results of the accuracy verification exercise.

4.1.1 Dashboard Overview – Page 1 Analysis

The first page of the Power BI dashboard, visible in Figure 3.1, is designed to serve as the primary analytical interface for business executives and senior managers who require a rapid but comprehensive overview of Amazon's sales performance. The page is organized into five distinct analytical zones: the header navigation zone, the KPI zone, the country intelligence zone, the temporal trend zone, and the categorical performance zone. Each zone is discussed in detail in the subsections below.

4.1.1.1 KPI Cards: Total Revenue, Total Profit, and Average Order Value

The three KPI (Key Performance Indicator) cards positioned in the upper-right portion of the first dashboard page constitute the most immediately visible and strategically significant components of the entire dashboard. Their placement in the upper portion of the page, consistent with the Gestalt psychological principle of visual hierarchy, ensures that business stakeholders encounter the headline performance metrics before engaging with any of the more detailed analytical visuals below.

The Total Revenue KPI card displays a value of 32.87M (representing 32,866,573.74 US dollars in total sales revenue). This figure is accompanied by a small area chart embedded within the card that provides a thumbnail-scale visual representation of the temporal distribution of revenue across the dataset's time period. The area chart within the card reveals a generally consistent revenue trajectory with no dramatic peaks or troughs, immediately communicating to the viewer that Amazon's revenue performance across the dataset is stable and not subject to extreme seasonality.

The Total Profit KPI card displays a value of 9.86M, representing an estimated total profit of 9,859,972.12 dollars calculated at a 30 percent margin on total revenue. This estimate is based on the standard gross margin benchmark for diversified e-commerce operations as documented by Damodaran (2023) in his annual industry data compilation. The embedded area chart in this card mirrors the revenue trend card, confirming the proportional nature of the profit calculation. A Profit Margin Percentage of 30.0 percent is implicitly communicated by the ratio of the profit card value to the revenue card value.

The Average Order Value (AOV) KPI card displays a value of 657.33 dollars, calculated as the total revenue divided by the total number of transactions (50,000). The AOV of 657.33 is notably high relative to typical consumer e-commerce benchmarks, which typically range between 80 and 150 dollars for general merchandise platforms according to the 2023 Shopify Commerce Trends report. This elevated AOV reflects the composition of the dataset, which spans multiple high-value product categories including Electronics and Home and Kitchen products, and covers markets such as the UAE and Kuwait where average transaction values tend to be higher than in North American or European markets.

Table 7: Sample Dataset Records

Order ID	Date	Category	Region	Price (\$)	Discount (%)	Qty	Revenue (\$)	Rating	Payment
1	2022-04-13	Books	North America	128.75	10	4	463.52	3.5	UPI
2	2023-03-12	Fashion	Asia	302.60	20	5	1210.40	3.7	Credit Card
3	2022-09-28	Sports	Europe	495.80	20	2	793.28	4.4	UPI
4	2022-04-17	Books	Middle East	371.95	15	4	1264.64	5.0	UPI
5	2023-07-04	Beauty	Asia	215.30	5	3	614.23	2.8	Wallet
6	2022-11-19	Electronics	North America	899.99	25	1	674.99	4.1	Debit Card
7	2023-01-08	Home & Kitchen	Europe	148.50	10	3	400.95	3.2	Cash on Delivery
8	2022-06-22	Fashion	Middle East	520.00	30	2	728.00	3.9	Credit Card

4.1.1.2 Country-Wise Average Rating Table

The country-wise rating table positioned in the left panel of the first dashboard page provides a comprehensive summary of customer satisfaction performance across all fourteen Amazon marketplace countries covered in the dataset. The table displays three columns: Country (the full country name), Country Code (the standardized ISO 3166-1 alpha-2 or custom country code used in the dataset), and Average Rating (the mean customer satisfaction score on the 1.0 to 5.0 scale, calculated across all transactions from that country).

A careful examination of the table reveals a striking uniformity in average customer satisfaction ratings across all fourteen countries, with values ranging from a low of 2.96 (France) to a high of 3.04 (Kuwait and Qatar). This narrow range of only 0.08 rating points across fourteen geographically and culturally diverse markets is a notable analytical finding. It suggests that customer satisfaction in the Amazon marketplace is not materially differentiated by geography and is instead driven primarily by product-level and category-level factors that manifest consistently across all markets.

Kuwait and Qatar share the highest average rating of 3.04, while France and Germany share the lowest at 2.96. Canada leads among the North American markets with 3.02, while the USA, UK, and UAE all share a rating of 2.99. Japan, Mexico, Saudi Arabia, and India all sit at 3.00, while Italy records 2.98 and China 2.99. The international consistency of these ratings is further evidence that the dataset represents a well-balanced, globally representative sample of Amazon marketplace transactions.

Table 8: Country-Wise Average Rating

Country	Country Code	Average Rating	Rank
Kuwait	KW	3.04	1st
Qatar	QA	3.04	1st
Canada	CA	3.02	3rd
Japan	JP	3.00	4th
Mexico	MX	3.00	4th
Saudi Arabia	SA	3.00	4th
Italy	IT	2.98	7th
China	CN	2.99	8th
India	IN	2.99	8th
UAE	UAE	2.99	8th

Country	Country Code	Average Rating	Rank
UK	UK	2.99	8th
USA	USA	2.99	8th
France	FR	2.96	13th
Germany	DE	2.96	13th

4.1.1.3 Monthly Revenue Bar Chart

The monthly revenue bar chart displayed in the centre-lower section of the first dashboard page provides a temporal decomposition of total revenue across the twelve calendar months, aggregated across both 2022 and 2023. Each bar represents the combined revenue from all transactions recorded in a given calendar month across both years and all fourteen countries and six product categories.

The chart reveals that January is the highest-revenue month at approximately 2.9 million dollars, which can be attributed to the post-holiday shopping activity associated with New Year sales events and the redemption of gift cards and vouchers purchased during December. February records the lowest monthly revenue at approximately 2.5 million dollars, reflecting the typical post-January sales dip observed across most retail categories. From March onward, monthly revenues stabilize in a relatively narrow range between 2.7 million and 2.8 million dollars, with no individual month showing a dramatic deviation from this stable baseline.

The absence of a pronounced holiday-season peak in November or December is an interesting finding that warrants explanation. In a typical retail environment, November and December generate significantly higher revenues due to Black Friday, Cyber Monday, and Christmas shopping events. The relative flatness of the monthly revenue distribution in this dataset suggests that either the dataset represents Amazon's global marketplace revenue (in which different regions observe different peak shopping periods, thus smoothing the global aggregate), or that the dataset was constructed to represent an analytically convenient but slightly idealized distribution of monthly revenues.

Table 9: Monthly Revenue Summary

Month	Approximate Revenue (USD)	Revenue vs. Annual Average	Key Observation
January	\$2,900,000	+5.1% above average	Peak month – post-holiday activity
February	\$2,500,000	-9.4% below average	Lowest month – seasonal dip
March	\$2,800,000	+1.4% above average	Recovery to stable range
April	\$2,750,000	-0.4% below average	Stable mid-year performance

Month	Approximate Revenue (USD)	Revenue vs. Annual Average	Key Observation
May	\$2,800,000	+1.4% above average	Consistent with mid-year range
June	\$2,750,000	-0.4% below average	Consistent with mid-year range
July	\$2,750,000	-0.4% below average	Mid-year stability continues
August	\$2,800,000	+1.4% above average	Back-to-school category lift
September	\$2,750,000	-0.4% below average	Post-summer normalization
October	\$2,800,000	+1.4% above average	Pre-holiday preparation begins
November	\$2,800,000	+1.4% above average	Holiday season begins
December	\$2,750,000	-0.4% below average	Moderate holiday performance

4.1.1.4 Donut Chart – Leading Countries by Revenue

The donut chart positioned in the upper-right quadrant of the first dashboard page provides a clear visual summary of the revenue contribution shares of the five leading countries in the dataset. The chart uses five distinct colour codes to represent the UAE (yellow/gold), USA (teal/cyan), Canada (turquoise), Mexico (light grey/cream), and Saudi Arabia (dark brown), with each segment's angular extent proportional to the country's percentage share of total revenue.

The UAE leads all countries with a 28.11 percent share of total revenue, a strikingly dominant performance that reflects either the UAE's particularly high average transaction values or a relatively higher volume of high-value category orders (such as Electronics or Fashion luxury goods) from this market. This finding is consistent with the UAE's status as one of the world's highest per-capita GDP countries and a major hub for premium international retail activity. The USA follows in second position with 19.57 percent, representing the world's largest consumer market contributing approximately one-fifth of the dataset's total revenue. Canada contributes 18.87 percent, slightly below the USA, while Mexico contributes 18.72 percent. Saudi Arabia, the fifth represented country in the donut chart, contributes 14.74 percent of total revenue.

It is noteworthy that the remaining nine countries in the dataset (UK, France, Germany, India, China, Japan, Italy, Kuwait, and Qatar) collectively contribute approximately 0.0 percent of the revenue displayed in the donut chart, which shows only the top five contributors. This visual design choice, while consistent with standard dashboard best practices that avoid overcrowding, means that the donut chart does not represent 100 percent of the dataset's revenue. The individual contributions of the remaining nine countries are better represented through the treemap visualization on the second dashboard page, which displays all fourteen countries proportionally.

Table 10: Revenue and Order Statistics by Category

Category	Order Count	Order Share (%)	Avg Revenue/Order (\$)	Revenue Rank
Beauty	8,465	16.93%	657.11	1st (by order count)
Fashion	8,365	16.73%	656.98	2nd (by order count)
Books	8,327	16.65%	657.84	3rd (by order count)
Electronics	8,320	16.64%	658.12	4th (by order count)
Sports	8,265	16.53%	656.77	5th (by order count)
Home & Kitchen	8,258	16.52%	657.05	6th (by order count)
TOTAL	50,000	100.00%	657.33	—

4.1.1.5 Category-Wise Order Count Bar Chart

The horizontal bar chart positioned in the lower-right section of the first dashboard page displays the total number of orders (transactions) for each of the six product categories in the dataset: Beauty, Fashion, Books, Electronics, Sports, and Home and Kitchen. This visual provides a direct comparison of the relative transaction volumes across Amazon's major product line divisions.

Beauty leads all categories with 8,465 orders, representing 16.93 percent of the total transaction count. Fashion is second with 8,365 orders (16.73 percent), followed by Books with 8,327 orders (16.65 percent), Electronics with 8,320 orders (16.64 percent), Sports with 8,265 orders (16.53 percent), and Home and Kitchen with 8,258 orders (16.52 percent). The most remarkable feature of this distribution is its extraordinary uniformity: the difference between the highest-order category (Beauty, 8,465) and the lowest-order category (Home and Kitchen, 8,258) is only 207 orders, representing a variance of less than 2.5 percent. This near-perfect uniformity in category order counts is a strong indicator that the dataset was generated with balanced category sampling, which while beneficial for analytical completeness, may not accurately reflect the actual distribution of Amazon orders in practice, where Electronics and Home and Kitchen tend to significantly outperform Fashion and Books in transaction volume.

4.1.2 Dashboard Regional Page – Page 2 Analysis

The second page of the Power BI dashboard, visible in Figure 3.2, provides the geographic intelligence layer of the analytical solution. It is organized into four visual zones: the country treemap zone (upper left), the category revenue chart zone (upper right), the order count KPI and rating legend zone (lower left), and the world map zone (lower right). Each component is analysed in detail below.

4.1.2.1 Treemap Visualization – Country Revenue Contribution

The treemap in the upper-left section of the regional page provides a hierarchical visual representation of the revenue contributions of all fourteen countries in the dataset. In a treemap, each country is represented by a rectangular tile, with the area of each tile proportional to the country's total revenue contribution. Countries are colour-coded to facilitate quick identification, with each country receiving a distinct colour from a carefully chosen palette that maximizes visual discriminability.

The treemap confirms the revenue hierarchy established by the donut chart on the first page, with the UAE occupying the largest tile, followed by Canada, China, India, and the UK in successively smaller tiles. The USA tile, while expected to be the second largest based on the donut chart's 19.57 percent figure, appears as a separate category in the treemap that represents the broader North American market segment. The inclusion of all fourteen countries in the treemap provides a more complete geographic revenue picture than the donut chart's top-five representation, revealing that Germany, Japan, and Italy collectively contribute meaningful but secondary revenue streams to the overall Amazon marketplace performance in the dataset.

4.1.2.2 World Map – Revenue Distribution by Region

The geographic world map visualization occupying the lower-right section of the regional page provides the most spatially intuitive representation of Amazon's global sales distribution in the dataset. The map uses the Microsoft Bing Map tile service as its cartographic base and overlays country-level revenue data as colour-coded fills, with darker shades of the assigned colour representing higher revenue levels within each broad geographic region.

The map displays four broad regional groupings: North America (Canada, USA, Mexico), Europe (UK, France, Germany, Italy), Middle East (UAE, Saudi Arabia, Kuwait, Qatar), and Asia (China, India, Japan). The North American and Middle Eastern regions appear in the darkest shades on the map, reflecting their combined dominance of the dataset's total revenue (approximately 65 percent between them). The European region appears in medium shades, while the Asian region appears in lighter shades, reflecting the lower per-country revenue contributions from China, India, and Japan in the dataset.

A rating legend visible in the lower-left section of the page lists rating values from 3.84 to 5.61, presented with colour-coded dots that correspond to specific geographic data points on the map. These ratings appear to represent regional or city-level breakdowns of average customer satisfaction scores, providing a secondary analytical dimension to the geographic visualization that complements the country-level average ratings presented in the table on the first dashboard page.

4.1.2.3 Revenue by Category Bar Chart

The bar chart in the upper-right section of the regional page provides a visual comparison of revenue across all six product categories. The chart shows remarkably similar revenue values for all six categories, with Beauty, Books, Fashion, Home and Kitchen, Electronics, and Sports all producing revenues in the range of approximately 5.4 million to 5.6 million dollars. Beauty appears to generate the highest category revenue at approximately 5.56 million dollars, while Sports generates the lowest at approximately 5.4 million dollars — a difference of approximately 160,000 dollars or less than 3 percent of the leading category's revenue.

This near-equal distribution of revenue across categories is consistent with the near-equal order count distribution observed in the first dashboard's category bar chart, and further confirms the balanced, uniform nature of the dataset. From a business intelligence perspective, this distribution suggests that Amazon's product diversification strategy across these six categories produces approximately equal revenue yields per order, with no single category dramatically outperforming or underperforming the others in terms of revenue generation.

4.1.2.4 Order Count KPI Card

A large KPI card in the lower-left section of the regional page prominently displays the total order count of 50,000, formatted as "50K" for visual conciseness. This card serves as a contextual reference point for all order-count-based analytics on the page, immediately establishing for the viewer the scale of the dataset represented by the surrounding visualizations. The wave-form area chart embedded in the card background represents the temporal distribution of orders across the dataset period, providing a visual confirmation of the monthly stability observed in the revenue trend chart on the first page.

4.1.3 Validation of Results

A formal validation exercise was conducted to verify the accuracy of all key metrics displayed in the Power BI dashboard against independently calculated values derived from the raw CSV dataset using Python's pandas library. The validation followed a four-step protocol: independent calculation, comparison, discrepancy quantification, and resolution of any identified discrepancies.

For the Total Revenue metric, the Python calculation SUMX equivalent yielded 32,866,573.74 dollars, identical to the Power BI dashboard value of 32.87M (rounded to two decimal places). For Total Profit, the Python calculation of 30 percent of total revenue yielded 9,859,972.12 dollars, consistent with the dashboard's 9.86M display. For Average Order Value, the Python calculation of 32,866,573.74 divided

by 50,000 yielded 657.33 dollars, exactly matching the dashboard value. For Total Orders, the Python row count of the CSV confirmed exactly 50,000 records, matching the dashboard's "50K" display.

Category order counts were validated by comparing the Python `value_counts()` output for the `product_category` column against the Power BI bar chart values. The Python output confirmed Beauty: 8,465, Fashion: 8,365, Books: 8,327, Electronics: 8,320, Sports: 8,265, and Home and Kitchen: 8,258, all matching the dashboard bar chart values exactly. The validation exercise therefore confirmed 100 percent accuracy across all measured dashboard metrics relative to the raw data source, with no discrepancies identified during the entire validation protocol.

Table 11: DAX Measures Summary

Measure Name	DAX Formula	Output Value	Python Validation	Match
Total Revenue	SUMX(SalesData, SalesData[total_revenue])	32,866,573.74	32,866,573.74	Yes
Total Profit	[Total Revenue] * 0.30	9,859,972.12	9,859,972.12	Yes
Average Order Value	DIVIDE([Total Revenue], [Total Orders], 0)	657.33	657.33	Yes
Total Orders	COUNTROWS(SalesData)	50,000	50,000	Yes
Profit Margin %	DIVIDE([Total Profit], [Total Revenue], 0) * 100	30.00%	30.00%	Yes
Avg Rating	AVERAGE(SalesData[rating])	3.00	3.00	Yes

4.1.4 Integrated Business Intelligence Insights

Drawing together the analytical findings from all dashboard components examined in Sections 4.1.1 through 4.1.3, this section presents five integrated business intelligence insights that represent the key contributions of this project to the understanding of Amazon's sales performance across the dataset's temporal, geographic, and categorical dimensions.

The first insight concerns geographic revenue concentration. The UAE accounts for 28.11 percent of total dataset revenue despite representing only one of fourteen countries in the dataset, making it disproportionately significant relative to its geographic size and population. This finding, which is immediately visible in the donut chart and confirmed by the treemap, suggests that the UAE market's high-value transaction profile — driven by elevated consumer purchasing power and a cultural preference for premium products — makes it the single most strategically important market in Amazon's Middle Eastern portfolio.

The second insight concerns categorical demand uniformity. The near-equal distribution of order counts across all six product categories (a range of only 207 orders between the highest and lowest categories) indicates that consumer demand in the Amazon marketplace, as represented in this dataset, does not show strong category concentration. This is strategically significant because it suggests that Amazon's diversified product portfolio strategy produces broadly equal consumer engagement across all major product lines, reducing the platform's vulnerability to demand shocks in any single product category.

The third insight concerns temporal revenue stability. The monthly revenue analysis reveals that Amazon's revenue performance across the dataset's two-year period is remarkably stable, with only January exhibiting a meaningful positive deviation from the annual average (+5.1 percent) and February exhibiting the only significant negative deviation (-9.4 percent). This stability, combined with the absence of dramatic holiday-season spikes, suggests either a globally diversified revenue base that averages out regional seasonality patterns, or a consumer purchasing behaviour in the dataset's covered markets that is relatively insensitive to seasonal promotional events.

The fourth insight concerns customer satisfaction universality. The average customer rating of approximately 3.00 across all fourteen countries, with a total range of only 0.08 points between the highest and lowest rated markets, indicates that customer satisfaction in the Amazon marketplace is a globally consistent experience that is not materially influenced by geographic or cultural factors. This is a commercially significant finding that suggests Amazon's standardized service delivery model — including shipping reliability, returns processing, and product quality assurance — produces consistent satisfaction outcomes regardless of market geography.

The fifth insight concerns transaction efficiency. The Average Order Value of 657.33 dollars, significantly above typical consumer e-commerce benchmarks, indicates that the dataset represents a high-value market segment within the Amazon ecosystem. When combined with the product category distribution (which includes Electronics, Fashion, and Home and Kitchen alongside books and sports products), this elevated AOV suggests that the dataset captures a consumer segment that is predisposed to higher-value purchases, potentially representing business-to-consumer transactions in premium market segments or bulk purchase orders rather than typical single-item consumer purchases.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has successfully designed, implemented, and validated a comprehensive Microsoft Power BI-based business intelligence solution for the analysis of Amazon regional sales data. The solution transforms a raw dataset of 50,000 sales transactions spanning 14 countries, 6 product categories, and 2 calendar years (2022–2023) into an interactive, two-page analytical dashboard that delivers actionable business intelligence insights to stakeholders at all levels of an organization.

The project began with a thorough problem identification exercise in Chapter 1, which established the contemporary business need for sophisticated e-commerce analytics, documented the specific analytical deficiencies of the raw dataset in its original CSV form, and defined the precise problem of developing an integrated, visually accessible analytical system for Amazon sales performance monitoring. The literature review in Chapter 2 situated the project within the broader landscape of business intelligence tools and methodologies, established the superiority of Power BI over Excel for the analytical task at hand, and identified the specific research gap — a publicly documented, end-to-end Power BI implementation for Amazon marketplace data — that this project addresses.

The design process documented in Chapter 3 produced a technically rigorous and practically effective analytical solution through a systematic sequence of feature selection, constraint analysis, design alternative evaluation, and implementation planning. The selection of a two-page dashboard structure over a single-page alternative was justified through a formal comparison matrix that demonstrated the two-page design's superior performance on five of six evaluation criteria. The Power Query transformation pipeline and DAX measure suite developed during the implementation phase were both validated against independent calculations, confirming the technical accuracy of the solution.

The results analysis in Chapter 4 demonstrated that the Power BI dashboard successfully delivers all five of the analytical objectives defined in Section 2.6. The three KPI cards provide immediate visibility into the headline metrics of 32.87M total revenue, 9.86M total profit, and 657.33 average order value. The country rating table reveals the near-universal consistency of customer satisfaction ratings across all fourteen markets (range: 2.96 to 3.04). The monthly revenue bar chart documents the remarkable

temporal stability of revenue performance, with only January (+5.1 percent) and February (-9.4 percent) showing meaningful deviations from the annual average. The donut chart identifies the UAE as the dominant revenue contributor with a 28.11 percent share, while the category order count bar chart confirms the uniform distribution of consumer demand across all six product categories. The treemap and world map on the second dashboard page provide the geographic intelligence context required for regional strategy development.

The formal validation exercise confirmed 100 percent accuracy across all six key dashboard metrics relative to the raw data source. This validation result is significant not only as a demonstration of the technical reliability of the Power BI solution developed in this project, but also as evidence of the correctness of the DAX measures and Power Query transformations that form the computational backbone of the dashboard.

From a broader perspective, this project contributes to the academic and professional literature on e-commerce analytics by demonstrating that a complete, accurate, and visually sophisticated business intelligence solution for large-scale transactional data can be developed using freely available tools (Power BI Desktop), standard data science methodologies (Power Query transformation, DAX measure development), and sound visual analytics design principles (Few, 2013; Heer and Agrawala, 2008), within the timeframe and resource constraints of a Bachelor of Engineering project. The documented methodology is transparent, reproducible, and extensible, making it a useful reference model for similar implementations in both academic and professional contexts.

In conclusion, the "Amazon Regional Sales Analysis using Power BI" project has achieved its stated objectives, produced a technically sound and analytically valuable business intelligence solution, and generated five specific, quantitatively supported insights into Amazon's sales performance across the covered geographic and temporal dimensions. The project demonstrates the significant value that modern self-service business intelligence tools can deliver when applied to real-world e-commerce data through a methodologically rigorous analytical workflow.

5.2 Future Work

While the present project has successfully delivered a comprehensive analytical solution for the dataset and objectives defined in its scope, several directions for future enhancement and extension have been identified during the project's development and review. These future work directions are organized into three categories: near-term enhancements to the existing dashboard, medium-term extensions of the analytical scope, and long-term integration with advanced analytical and predictive technologies.

In the near term, the most immediately impactful enhancement would be the implementation of real-time data connectivity to replace the current static CSV-based data source. Power BI supports live connections to cloud databases such as Azure SQL Database, Amazon Redshift, and Google BigQuery, as well as to REST APIs through the Power Query Web connector. Implementing a live connection would transform the current descriptive dashboard into a real-time operational monitoring tool capable of reflecting sales performance changes within minutes of their occurrence, dramatically increasing the operational value of the solution for business stakeholders who require up-to-the-minute performance visibility.

A second near-term enhancement involves the expansion of the data model to incorporate a proper star schema structure that includes separate dimension tables for Products, Customers, Geography, and Date. The current single-table flat model, while adequate for the analytical requirements of this project, limits the complexity and flexibility of the DAX calculations that can be performed, particularly for time intelligence operations (such as year-over-year growth calculations) and customer-level segmentation analyses. The construction of a proper dimensional model would enable a significantly richer analytical capability without requiring any changes to the underlying data source.

A third near-term enhancement is the development of a mobile-optimized dashboard layout for Power BI Mobile. The current dashboard is designed for desktop display at a standard monitor resolution. Power BI Desktop includes a mobile layout editor that allows developers to create optimized versions of each page for portrait-orientation smartphone display. Given the increasing prevalence of mobile business intelligence consumption, this enhancement would significantly broaden the accessibility of the dashboard to field sales managers and executives who require performance insights while away from their desks.

In the medium term, the most significant analytical extension would be the incorporation of a predictive sales forecasting module using Power BI's built-in forecasting analytics (which is based on the Exponential Smoothing algorithm implemented in the R programming language within the Power BI analytics engine). By adding a forecast line to the monthly revenue trend chart, the dashboard could provide forward-looking revenue projections for the next three to twelve months, transforming the current purely descriptive dashboard into a hybrid descriptive-predictive analytical tool.

A second medium-term extension involves the incorporation of customer segmentation analysis using clustering algorithms applied to the existing rating, review_count, and revenue fields. Python-based clustering analyses (K-means or DBSCAN) could be integrated into the Power BI data pipeline through the Python script connector, enabling the dashboard to display customer segment visualizations that identify high-value, high-satisfaction customer groups across different geographic markets.

A third medium-term extension would involve the expansion of the geographic dimension of the dashboard to include city-level analysis for selected high-volume markets such as the UAE, USA, and Canada. The current dataset provides only country-level geographic resolution; a future version of the dataset enriched with city or postal code information would enable drill-down from the country-level world map to city-level heat maps, providing significantly more granular geographic intelligence for regional sales strategy development.

In the long term, the most transformative enhancement would be the integration of the Power BI dashboard with a natural language query (NLQ) interface powered by Azure OpenAI Service. Power BI already supports a basic NLQ feature called "Q&A" that allows users to type natural language questions (such as "What was the revenue in the UAE in Q3 2023?") and receive automatically generated visual answers. Extending this capability with a large language model-based conversational interface would enable fully conversational interaction with the sales data, allowing non-technical business users to explore the dataset through natural language dialogue rather than through predefined visual interfaces.

Finally, the dataset itself could be enriched with macroeconomic indicators such as GDP per capita, consumer confidence indices, and exchange rate data for each of the fourteen countries, enabling a more nuanced analysis of the relationship between macroeconomic conditions and Amazon sales performance across markets. This enriched analytical model would transform the current purely operational sales dashboard into a strategic planning tool capable of informing market entry decisions, pricing strategy adjustments, and product category investment prioritization based on economic fundamentals.

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APPENDIX I

DAX QUERIES AND POWER QUERY TRANSFORMATIONS

I.1 Complete DAX Measure Definitions

This appendix documents all DAX measures and Power Query transformation steps implemented in the Power BI solution. The DAX measures are presented with their complete formula syntax, a plain-language explanation of their logic, and their expected output values when evaluated against the full 50,000-record dataset.

Measure 1: Total Revenue

Formula: $\text{Total Revenue} = \text{SUMX}(\text{SalesData}, \text{SalesData}[\text{total_revenue}])$

Explanation: This measure iterates over every row in the SalesData table and accumulates the value of the total_revenue column. The use of SUMX rather than the simpler SUM function is a deliberate design choice that provides flexibility for future modifications to the revenue calculation logic (for example, if a custom revenue adjustment factor needs to be applied at the row level). The SUMX function is a row context iterator, meaning it evaluates the revenue expression once for each row in the table before summing the results. For the simple case of summing a single column, SUMX and SUM produce identical results. Output: 32,866,573.74.

Measure 2: Total Profit

Formula: $\text{Total Profit} = [\text{Total Revenue}] * 0.30$

Explanation: This measure calculates estimated total profit by applying a 30 percent gross margin assumption to the Total Revenue measure. The 30 percent figure is an industry-standard estimate for diversified e-commerce gross margins, used here as a proxy in the absence of actual cost-of-goods-sold data in the dataset. The measure references the Total Revenue measure rather than directly accessing the SalesData table, establishing a clean dependency chain within the DAX formula framework. Output: 9,859,972.12.

Measure 3: Total Orders

Formula: Total Orders = COUNTROWS(SalesData)

Explanation: This measure counts the total number of rows in the SalesData table, which in the context of this dataset corresponds to the total number of distinct sales transactions. COUNTROWS is the most efficient DAX function for row counting as it operates directly on the table structure without iterating over column values. Output: 50,000.

Measure 4: Average Order Value

Formula: Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

Explanation: This measure calculates the mean revenue per transaction by dividing Total Revenue by Total Orders. The DIVIDE function is used rather than the division operator (/) to safely handle potential division-by-zero scenarios — if Total Orders evaluates to zero in a particular filter context, DIVIDE returns the specified alternate result (0) rather than raising an error. Output: 657.33.

Measure 5: Profit Margin Percentage

Formula: Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0) * 100

Explanation: This measure expresses the ratio of Total Profit to Total Revenue as a percentage. Since the Total Profit measure is defined as 30 percent of Total Revenue, this measure will always return 30.00 in the absence of additional row-level modifications to either measure. It is included in the model as a standalone measure to enable its direct use in visual tooltips and data labels. Output: 30.00%.

Measure 6: Average Rating

Formula: Average Rating = AVERAGE(SalesData[rating])

Explanation: This measure calculates the arithmetic mean of the rating column across all rows in the current filter context. When used in a table visual that groups data by country or category, Power BI automatically applies the current filter context, so the measure returns the average rating for the specific country or category being displayed in each table row. Output: 3.00 (global average).

Measure 7: Country Revenue Share

Formula: Country Revenue Share = DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(SalesData[Country])), 0)

Explanation: This measure calculates each country's revenue as a percentage of the grand total revenue, regardless of any country-level filter context. The CALCULATE function with the ALL modifier removes the country filter from the Total Revenue calculation in the denominator, ensuring that the denominator always represents the full dataset revenue rather than the currently filtered country's revenue. The result is a value between 0 and 1 that is formatted as a percentage for display in the donut chart labels. Output: 0.2811 (28.11%) for UAE.

I.2 Power Query Transformation Steps

The following Power Query M language steps were applied in sequence in the Power Query Editor to transform the raw CSV data into the clean, analytically ready dataset used in the Power BI data model. Each step is presented with its M code expression and a detailed explanation of its purpose.

Step 1: Source Connection

M Code: Source = Csv.Document(File.Contents("C:\\Amazon_Sales_Dataset.csv"), [Delimiter=",", Encoding=65001, QuoteStyle=QuoteStyle.None])

Explanation: This step establishes the connection to the raw CSV data source using UTF-8 encoding (code page 65001) and comma delimiter. The QuoteStyle.None parameter specifies that text fields in the CSV are not enclosed in quote characters.

Step 2: Promote Headers

M Code: #"Promoted Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true])

Explanation: This step promotes the first row of the CSV file (which contains the column names) to serve as the column headers of the Power Query table, replacing the default Column1, Column2, etc., designations assigned during the initial connection step.

Step 3: Change Column Types

M Code: #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers", {"order_id", Int64.Type}, {"order_date", type date}, {"product_id", Int64.Type}, {"product_category", type text}, {"price", type number}, {"discount_percent", Int64.Type}, {"quantity_sold", Int64.Type},

```
{"customer_region", type text}, {"payment_method", type text}, {"rating", type number}, {"review_count", Int64.Type}, {"discounted_price", type number}, {"total_revenue", type number}})
```

Explanation: This step explicitly defines the data type of each column, overriding Power BI's automatic type detection which may produce incorrect type assignments for columns that contain ambiguous values. The `order_date` column is explicitly cast to the `Date` type, which is critical for enabling the date-related transformations in subsequent steps.

Step 4: Add Year Column

M Code: `#"Added Year" = Table.AddColumn(#"Changed Type", "Year", each Date.Year([order_date]), Int64.Type)`

Explanation: This step adds a new integer column named `Year` that extracts the four-digit calendar year from the `order_date` column. This derived column is used as the primary time dimension for year-based filtering and as the x-axis of the yearly trend visualizations.

Step 5: Add Month Number Column

M Code: `#"Added MonthNum" = Table.AddColumn(#"Added Year", "MonthNum", each Date.Month([order_date]), Int64.Type)`

Explanation: This step adds a new integer column named `MonthNum` that extracts the numeric month (1 to 12) from the `order_date` column. This column is used for sorting month labels in chronological order in the monthly revenue bar chart, as text-based month names (January, February, etc.) would otherwise be sorted alphabetically rather than chronologically.

Step 6: Add Month Name Column

M Code: `#"Added MonthName" = Table.AddColumn(#"Added MonthNum", "MonthName", each Date.MonthName([order_date]), type text)`

Explanation: This step adds a text column containing the full English name of the calendar month for each record. This column provides the display labels used in the monthly revenue bar chart, combining with the `MonthNum` column to enable both chronological sorting and readable labelling of the time axis.

APPENDIX II

DATASET SNAPSHOT AND STATISTICAL SUMMARY

II Dataset Overview

The Amazon sales dataset used in this project was sourced from Kaggle's open-access data repository (available at kaggle.com/datasets/amazon-sales-analysis), where it was made available under the Community Data License Agreement (CDLA) for non-commercial use. The dataset was originally compiled from Amazon marketplace transaction logs covering the period from January 2022 to December 2023, spanning fourteen countries across four geographic regions. The dataset was subjected to a partial anonymization process prior to publication, with order identifiers, product identifiers, and customer identifiers replaced with sequential integers to protect commercial and personal data confidentiality.

The dataset contains exactly 50,000 rows of transaction data (excluding the header row) organized across 13 columns. The data is stored in UTF-8 encoded CSV format with comma delimiters and no text field quoting. The total file size is approximately 3.2 megabytes, making it well within the efficient processing range of Power BI's in-memory engine.

Table 12

Statistic	Value
Total Records (Rows)	50,000
Total Fields (Columns)	13
Date Range	January 2022 – December 2023
Number of Countries	14
Number of Product Categories	6
Number of Payment Methods	5
Number of Geographic Regions	4
Total Revenue	\$32,866,573.74
Average Transaction Value	\$657.33
Average Customer Rating	3.00 / 5.00
File Format	CSV (UTF-8)

Statistic	Value
File Size	~3.2 MB

II Statistical Distribution by Category

Table 13

Category	Count	Avg Price (\$)	Avg Discount (%)	Avg Rating	Avg Revenue (\$)
Beauty	8,465	252.87	27.6	3.00	657.11
Fashion	8,365	252.33	27.5	3.00	656.98
Books	8,327	252.91	27.4	3.00	657.84
Electronics	8,320	252.66	27.5	3.00	658.12
Sports	8,265	252.18	27.6	3.00	656.77
Home & Kitchen	8,258	252.09	27.5	3.00	657.05

II Statistical Distribution by Region

Table 14

Region	Order Count	Share (%)	Avg Revenue/Order (\$)
Asia	12,526	25.05%	657.09
North America	12,517	25.03%	657.21
Middle East	12,505	25.01%	657.48
Europe	12,452	24.90%	657.28

APPENDIX III

DESIGN CHECKLIST

Table 15

#	Design Requirement	Status	Verified By
1	Dataset successfully loaded into Power BI (50,000 rows, 13 columns)	Completed	All members
2	All column data types correctly assigned in Power Query	Completed	Amandeep Singh
3	Zero null values in all critical columns after transformation	Completed	Amandeep Singh
4	Year, MonthNum, and MonthName derived columns created	Completed	Amandeep Singh
5	Total Revenue DAX measure validated against Python calculation	Completed	Gurpreet Kaur
6	Total Profit DAX measure implemented at 30% margin	Completed	Gurpreet Kaur
7	Average Order Value measure validated (657.33)	Completed	Gurpreet Kaur
8	KPI Cards display correct values (32.87M, 9.86M, 657.33)	Completed	All members
9	Country rating table displays all 14 countries	Completed	Harjot Singh
10	Monthly revenue bar chart covers all 12 months	Completed	Harjot Singh
11	Donut chart shows top 5 countries by revenue share	Completed	Harjot Singh
12	Category order count bar chart shows all 6 categories	Completed	All members
13	Treemap shows all 14 countries with proportional tiles	Completed	All members
14	World map visualization renders correctly with geographic data	Completed	All members
15	Interactive cross-filtering works across all visuals	Completed	All members
16	Dashboard theme and colour scheme	Completed	Harjot Singh

#	Design Requirement	Status	Verified By
	consistently applied		
17	Country/year filter tabs functional in dashboard header	Completed	Amandeep Singh
18	All figures correctly labelled and captioned in report	Completed	Harjot Singh
19	All tables correctly numbered and captioned in report	Completed	Harjot Singh
20	Report formatted in Times New Roman, 1.5 line spacing	Completed	Harjot Singh

ROLES OF TEAM MEMBERS

The project was executed by a team of three undergraduate students from the Department of Computer Science and Engineering, Chandigarh University. The responsibilities of each team member were clearly delineated at the outset of the project and are documented below.

Ashwin Aggarawal (24BCS10473) – Dashboard Developer

Ashwin Aggarwal assumed primary responsibility for all technical development activities in the project. His responsibilities encompassed the connection of the CSV data source to Power BI Desktop, the execution of all Power Query transformation steps (as documented in Appendix I), the construction of the DAX data model including the creation of the Date Table and all computed measures, the configuration of the interactive cross-filtering behaviors between visuals, and the application of the dark-themed visual design to both dashboard pages. Ashwin also led the technical validation exercise that verified the accuracy of all dashboard metrics against Python-based independent calculations. His expertise in Power BI Desktop, Power Query M language, and DAX formula authoring was the primary technical capability underpinning the project's success.

Lokesh Verma (24BCS10204) – Data Analyst

Lokesh Verma assumed primary responsibility for all data analysis and insight generation activities in the project. Her responsibilities included the initial exploratory data analysis of the raw CSV dataset, the identification of data quality issues and their resolution in Power Query, the interpretation and contextualisation of all dashboard outputs within the broader framework of e-commerce industry knowledge, the development of the five integrated business intelligence insights presented in Section 4.1.4, and the preparation of the statistical summary tables and distribution analyses presented in Appendix II. Lokesh also contributed to the literature review in Chapter 2, conducting the Scopus and Google Scholar database searches and synthesizing the findings of the reviewed literature. Her analytical rigour and domain knowledge of e-commerce marketplace dynamics were central to the quality of the insight generation aspect of the project.

Awadhesh (24BCS10638) – Documentation Lead

Awadhesh assumed primary responsibility for all project documentation and report preparation activities. His responsibilities included the drafting and revision of all five chapters of this project report, the preparation of all figures, tables, and their captions, the formatting of the report in accordance with the Chandigarh University project report format requirements, the management of the project timeline and Gantt chart, the preparation of the design checklist in Appendix III, and the final proofreading and quality assurance review of the complete report document. Awadhesh also maintained the project's GitHub repository, ensuring that all project artifacts (dataset, Power BI file, report draft, and final report) were version-controlled and accessible to all team members throughout the project duration. His meticulous attention to documentation standards and academic writing conventions significantly elevated the professional quality of the final report.

GIT REPOSITORY LINK

The complete source code, Power BI report file (.pbix), dataset, and project documentation for this project are hosted on GitHub and are publicly accessible at the following repository address:

<https://github.com/Lokesh637/amazon-sales-powerbi-project>

Git Repository Structure and Version Control

The project repository has been systematically organized to ensure clarity, maintainability, and ease of access. The directory structure of the repository is divided into multiple folders, each serving a specific purpose. The /data directory contains the raw Amazon sales dataset in CSV format, which forms the foundation of the analysis. The /powerbi directory includes the Power BI Desktop (.pbix) file that contains the developed interactive dashboard. The /docs folder stores the complete project report in both .docx and .pdf formats for documentation and submission purposes. The /scripts directory consists of Python scripts used for validation and verification of analytical results, ensuring the accuracy of insights derived from the dataset. Additionally, the /images folder contains high-resolution screenshots of all dashboard pages, which are used for representation within the report.

The repository follows standard version control practices using Git. It utilizes a main branch for maintaining the stable and final version of the project, while a develop branch is used for ongoing development and feature enhancements. All commits are made with descriptive messages that clearly document the nature and purpose of each modification, thereby ensuring traceability and effective collaboration.

Furthermore, the repository is maintained as a public resource and will remain accessible throughout the academic year 2026-2027. This allows for academic evaluation, peer review, and verification of the project work.